CLMM
Online Meta-Learning: Summary
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Online Meta-Learning

Motivation. Two distinct research paradigms address how agents can leverage past experience to improve future learning. Meta-learning studies how to learn a prior over model parameters for fast adaptation, but assumes tasks are available as a batch. Online (regret-based) learning handles sequential, non-stationary task streams, but trains a single model without task-specific adaptation.

Neither setting alone captures continual lifelong learning: meta-learning neglects sequentiality, and online learning does not consider how past experience can accelerate adaptation to new tasks.

Core Idea. The paper introduces the **online meta-learning** problem setting, which merges ideas from both paradigms. An agent faces tasks one after another in a sequential fashion and must simultaneously:

- Use past experience to learn good priors (meta-learning).
- Adapt quickly to the current task at hand (online learning).

At each round t, the agent maintains parameters $\mathbf{w}_t$, the world reveals a task defined by loss f_t, and the agent applies a local update procedure U_t (e.g., a gradient descent step) before being evaluated.

Follow the Meta Leader (FTML). The proposed algorithm extends MAML to the online setting by applying the Follow the Leader (FTL) principle at the meta-level. FTML selects parameters at each round by solving:

$$\mathbf{w}_{t+1} = \arg\min_{\mathbf{w}} \sum_{k=1}^{t} f_k\big(U_k(\mathbf{w})\big).$$

This can be interpreted as playing the best meta-learner in hindsight up to round t. The comparator class is powerful: it allows task-specific adaptation via U_t, so achieving sublinear regret means the algorithm is competitive with the best MAML-style initialization in hindsight.

Theoretical Guarantees. Under standard smoothness assumptions plus one additional condition on Lipschitz Hessians (C^2-smoothness), the authors prove:

(a) The MAML-like objective inherits convexity from the original loss when the step size α is sufficiently small (Theorem 1).

(b) FTML achieves $O\left(\frac{G^2}{\mu}\log T\right)$ regret against the best meta-learner in hindsight (Corollary 2), providing the first provable and efficient guarantees for MAML-like objectives.

Practical Algorithm. For deep networks with non-convex losses, the paper adapts FTML using stochastic optimization. At each round, the algorithm maintains a task buffer B and computes meta-gradients by sampling past tasks and using independently drawn minibatches for the inner update ($\mathcal{D}_k^{\text{tr}}$) and outer optimization ($\mathcal{D}_k^{\text{val}}$) to reduce bias.

The gradient computation follows:

$$\mathbf{g}_t = \nabla_{\mathbf{w}} \, \mathbb{E}_{k \sim \nu^t} \mathcal{L}\big(\mathcal{D}_k^{\text{val}}, U_k(\mathbf{w})\big), \quad \text{where} \quad U_k(\mathbf{w}) \equiv \mathbf{w} - \alpha \, \nabla_{\mathbf{w}} \mathcal{L}(\mathcal{D}_k^{\text{tr}}, \mathbf{w}).$$

Experimental Results. FTML is evaluated on three vision-based sequential learning problems:

- **Rainbow MNIST:** 56 tasks with varying backgrounds, scales, and rotations. FTML learns new tasks faster over time and outperforms all baselines in both efficiency and final performance.
- **Five-Way CIFAR-100:** A sequence of 5-way classification tasks. FTML learns each task more efficiently than training from scratch and benefits from adapting all layers rather than only the last layer.
- **PASCAL 3D+ Pose Prediction:** 90 pose regression tasks. FTML solves many tasks with only 10 datapoints, demonstrating strong forward transfer.

Across all experiments, FTML substantially outperforms baselines including Train on Everything (TOE), training from scratch, and FTL with fine-tuning.

Limitations. FTML inherits the computational cost of FTL, which grows as loss functions accumulate. Storing all datapoints from prior tasks may also be impractical at scale. The theoretical analysis is restricted to convex losses with C^2-smoothness, leaving non-convex guarantees as future work.

Takeaway. Online Meta-Learning bridges meta-learning and online learning into a unified framework for continual lifelong learning. The FTML algorithm demonstrates that meta-learning sequentially—becoming more proficient at learning new tasks over time—is both theoretically grounded and empirically effective.